

ANH NGUYEN HOANG

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RESEARCH INTERESTS

My current research focuses on AI for Software Engineering, particularly foundation models for code, autonomous coding agents, and long-horizon software evolution. I aim to study principled approaches for training and adapting code-focused language models, as well as designing reliable agentic systems for repository-scale software engineering tasks. I am also open to diversifying my research across related areas in the future.

EDUCATION

Hanoi University of Science and Technology

2020 – 2024

Bachelor of Science in Computer Science

GPA: 3.75/4.0

Ranked 15th of 296 students (Top 5%)

PUBLICATIONS

** denotes equal contribution.*

- [1] **CodeWiki: Evaluating AI’s Ability to Generate Holistic Documentation for Large-Scale Codebases**

[\[Paper\]](#) [\[Code\]](#)

Anh Nguyen Hoang, Minh Le-Anh, Bach Le, Nghi D. Q. Bui

Findings of ACL 2026

- [2] **Mutual-Pairing Data Augmentation for Few-Shot Continual Relation Extraction** [\[Paper\]](#) [\[Code\]](#)

Nguyen Hoang Anh*, Quyen Tran*, Thanh Xuan Nguyen*, Nguyen Thi Ngoc Diep, Linh Ngo Van, Thien Huu Nguyen, Trung Le

NAACL 2025

- [3] **Preserving Generalization of Language Models in Few-Shot Continual Relation Extraction** [\[Paper\]](#)

[\[Code\]](#)

Quyen Tran*, Nguyen Xuan Thanh*, **Nguyen Hoang Anh***, Nam Le Hai, Trung Le, Linh Van Ngo, Thien Huu Nguyen

EMNLP 2024

PREPRINTS

- [1] **RustPrint: Documentation-Guided Agentic Codebase Migration from C to Rust** [\[Preprint\]](#)

Minh Le-Anh, **Anh Nguyen Hoang**, Bach Le, Nghi D. Q. Bui

Preprint, 2026

HONORS AND AWARDS

Champion, AI for Real Action 2025

January 2026

First prize with Team Converged among nearly 400 teams in Vietnam’s first national artificial intelligence competition

- **Recipient, Techcombank Overseas PhD Study Sponsorship**

April 2026

Sponsorship limit of USD 333,000 for overseas PhD study [\[Offer Letter\]](#)

EXPERIENCE

FPT Software AI Center

May 2025 – Present

AI Research Resident

Hanoi, Vietnam

Advisors: Associate Prof. Bach Le, University of Melbourne; Assistant Prof. Nghi D. Q. Bui, VinUniversity

- Developed CodeWiki, an LLM-based framework that generates holistic, repository-level documentation for large software projects
- Research documentation-guided code migration, automated program repair, vulnerability detection, and secure coding agents

EXPERIENCE (CONTINUED)

Research Assistant

March 2024 – April 2025

Continual Learning and Natural Language Processing Research

Hanoi, Vietnam

Advisors: Assistant Prof. Linh Ngo Van; Associate Prof. Thien Huu Nguyen

- Researched few-shot continual relation extraction and generalization, resulting in equal-contribution papers at EMNLP 2024 and NAACL 2025

Oraichain Labs

May 2024 – April 2025

AI Engineer

Hanoi, Vietnam

- Led the development of Cupiee, an emotional AI companion platform with more than 300,000 users and approximately 100,000 messages per day
- Designed autonomous multi-agent systems for social media engagement and market analysis

ONGOING RESEARCH PROJECTS

CHAIR: Hypothesis-Driven Agent-Harness Optimization

2026 – Present

Research Lead; advised by Associate Prof. Bach Le and Assistant Prof. Nghi D. Q. Bui, FPT Software AI Center

- Developing a research-process controller that improves agent harnesses through causal hypothesis generation, experiment selection based on information gain, and controlled single-factor evaluation

CGSA: Coverage-Guided Agentic Static Analysis

2026 – Present

Collaborator; led by Prof. Lingming Zhang’s group at the University of Illinois Urbana-Champaign

- Contributing to an agentic vulnerability-detection framework that uses previously fixed vulnerabilities as reference cases to validate findings and reduce false positives

MCPGuard: Fine-Tuning Secure Coding Agents against MCP Tool Poisoning

2026 – Present

Collaborator; led by researchers at Mila – Quebec AI Institute

- Contributing to the design of adaptive tool-poisoning attacks that generate diverse attack trajectories and reduce overfitting to fixed attack templates during secure-agent training

SELECTED OPEN-SOURCE SOFTWARE

CodeWiki

1,500+ GitHub stars

Open-source repository documentation framework supporting seven programming languages

XTTSv2-Finetuning-for-New-Languages

200+ GitHub stars

Toolkit for adapting the XTTSv2 speech-synthesis model to new languages

LANGUAGES

Vietnamese

Native proficiency

English

Professional working proficiency (IELTS 7.0)

REFERENCES

Associate Prof. Bach Le

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Assistant Prof. Nghi D. Q. Bui

Staff Research Scientist, Google Research

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Assistant Prof. Linh Ngo Van

Hanoi University of Science and Technology, Vietnam

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Associate Prof. Thien Huu Nguyen

University of Oregon, USA

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